

CS430 Final Project: Spending Tracker

Cameron Bradley Jack Duensing

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Introduction

Purpose

Our goal when creating this project was to make a spending tracking tool we would actually use. This web app is a place where purchases can be stored, recalled, and analyzed. We didn't like modern budgeting apps with the extra fluff of telling you where and how your money should be spent. The idea of something smaller and specialized created this spending tracker, where the only goal is to store where your money goes. We track what you spent money on categorically, as well as the amount, the location, the time and date, the medium of payment, and whether or not it was spent online. This allows for analytics to be generated that show your spending habits in a digestible format, while keeping data entry simple.

Scope

The scope of this project is intentionally narrow. We did not intend to compete with bigger budgeting apps who have more features and better analytics. Our sole goal was to create a centralized ledger and analysis tool to show you where your money ends up. This required the creation of one tool, the web app itself. Included in this are portions to log in, enter new purchases, and look at analytics for past purchases.

Details

This project has a relatively straightforward 'how'. It is a server hosted web application, hosted in our basement on Ubuntu server. The app uses Flask for the web framework to keep everything simple and light. The backend database is hosted on Microsoft SQL Server 2025. Plotly was used for the creation of all the graphs and charts. Implementation of the login system is done with cookies, as modern pages do.

Contributions

Cameron

- Frontend
 - HTML
 - CSS
 - JS
 - Password authentication system
 - Flask implementation
 - Forms for data collection
 - Graphical design
 - CrewAI Integration

Jack

- Backend
 - Database schema
 - Database setup
 - Fields engineering/ERD generation
 - SQL queries for sending data to frontend
 - Graph creation with plotly
 - Creation of synthetic data

Usage

Launch

This is a server hosted web app so the only requirements to launch are a modern browser. The user simply navigates to the address and the app is presented.

Features

- Secure log in system using cookies
- Password changing
- Add new purchases
- See previous 5 purchases
- See spending over time as a stacked area chart

- See your purchases by subcategory as a percentage of total purchases and amount spent
- See where you spend your money with a map chart

Technical Breakdown

Component Implementation

Frontend The frontend of this project has many components that make it natural and intuitive for users. Starting at the highest level the main front facing pages are all built with HTML and CSS to create pages that are appealing to look at and self explanatory to use. HTML forms are the backbone of these pages, these forms allow for ease of transfer from the user to the system. The entries for these forms are collected using JavaScript. These forms include; login, creating users, resetting passwords, and entering purchases. Another component of the front end is the AI agent. This agent was implemented using CrewAI with a free Gemini model. The agent is available to be asked questions about purchase history and spending data. Finally, the largest feature of the frontend are the graphs and charts. These figures were created using plotly. The graphs include a table of your last 5 purchases, an area chart showing cumulative spending over different time frames, 2 pie charts that break down your purchases by count and amount, and finally a map chart that shows where your purchases were made. These figures are created on the backend are returned to the frontend using Flask.

Backend The backend of this project consists of three main components; the database, the data analytics, and the collection of data from the user. The data is collected from the user using python, this is done with the help of an Apache 2 server to collect, transform, and pipeline the data from the JavaScript to the backend. This also includes the processing of user requests, such as pulling the graphs. Next, the database is the backbone of this project. The database is ran as a Microsoft SQL Server 2025 database, locally hosted. The database runs on the database schema you will see below in Figure 1. This schema allows for separation of data into logical tables, but is related in a way that is conducive for constructing queries efficiently. Finally, the data analytics is the most important part of the project as it is the portion that shows the users their data. The figures are created using plotly express and plotly graph objects. The figures get their data from SQL queries, using pandas to read the queries into dataframes, occasionally some extra processing was required to get the data ready to be plotted. This included timeslicing and pivoting.

System Architecture The system is a locally hosted type 1 hypervisor running Ubuntu server on a virtual machine. The front end is facilitated with Flask, and the backend is hosted on Apache 2.

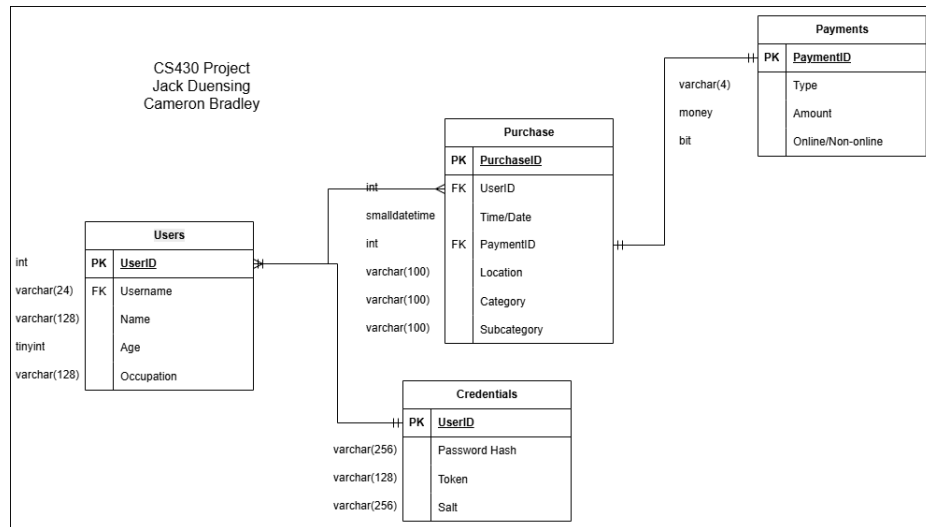


Figure 1: The ERD of the spending database

Packages/Tools

Languages

- HTML
 - Frontend bones
- CSS
 - Frontend styling
- JS
 - Adds front end dynamicness
- Python
 - Standard usage for data analytics
- SQL
 - Fetching database records

Packages

- Plotly/Plotly.js
 - Graph creation
- Pandas
 - Data analytics

- Hashlib
Password hashing
- Secrets
Password hashing
- Geopy
Pulling location data
- Datetime
Getting system time
- SQLAlchemy
Connecting to Microsoft SQL Server
- Fetch API
Asynchronous handling
- Flask
Webpage framework

Tools

- Proxmox
Hardware host: Type 1 Hypervisor
- VSCode
Commonplace tool for data analytics
- Notepad
Creation of HTML, CSS, JS
- Nano
Debugging on server
- Apache
Web server